PRE-ALGEBRA (7TH) · UNIT 4: FACTORS, FRACTIONS, AND EXPONENTS

Space Mission Packer

Use factors, fractions, exponents, and scientific notation to pack a mission to Mars.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Mission control has chosen your team to pack the cargo bay for a crewed mission to Mars. Supplies must be split into identical kits, cargo space is measured in fractions of the bay, and distances in space are huge.

You will use factors, fractions, exponents, and scientific notation to pack the ship and plan the journey.



Crew

6 astronauts

18-month mission

Distance

Mars ≈ 2.25×10^8 km

from Earth (average)

Learning Goals

- I can use prime factorization and the GCF to solve packing problems.
- I can simplify and compare fractions and identify rational numbers.
- I can use exponent rules and scientific notation.

Project Timeline

When	What you will do
Day 1	Supply Kits
Day 2	Cargo Fractions
Day 3	Exponents in Space
Final	Mission Packing Manifest + reflection

Part 1: Supply Kits

Day 1 · Use the Supply Cards.

Supply Cards

Water Pouches	Meal Packs	Oxygen Canisters
432 pouches	360 packs	96 canisters

Medical Kits	Seed Packets	Batteries
48 kits	144 packets	240 batteries

1. Write the prime factorization of 432, 360, and 144 using exponents.

2. Find the GCF of 432, 360, and 144. How many identical kits can be made with water, meals, and seeds? What is in each kit?

3. Can all six supplies go into the same identical kits? Find the GCF of all six amounts.

4. List what goes in each of the 24 kits.

Part 2: Cargo Fractions

Day 2 · The 24-section cargo bay holds the kits.

5. Each kit fills $\frac{1}{3}$ of a section. How many sections do 24 kits fill? What fraction of the bay is that?

6. Science equipment fills $\frac{5}{12}$ of the bay and spare parts fill $\frac{1}{6}$. Write the fraction of the bay that is still empty.

7. Classify each as rational or irrational: $\frac{5}{12}$, 0.333..., √2, π, √16.

8. Order $\frac{5}{12}$, 0.4, $\frac{3}{8}$, and $\frac{4}{9}$ from least to greatest.

Part 3: Exponents in Space

Day 3 · Use exponent rules and scientific notation.

9. Simplify $10^8 \times 10^3$ and $\frac{10^8}{10^5}$. Explain the rules.

10. Light travels about 3×10^5 km per second. About how many seconds does a radio message take to reach Mars at 2.25×10^8 km?

11. Mars has a diameter of 6.8×10^3 km and Earth 1.27×10^4 km. About how many times wider is Earth?

Final Product: Mission Packing Manifest

Your final product must include:

- Prime factorizations and GCF kit plan
- Contents of each kit
- Cargo bay fractions
- Exponent and scientific notation calculations
- Packing manifest and mission fact sheet

Reflection

R1. How does prime factorization make finding the GCF faster?

R2. Explain why $\frac{10^8}{10^5} = 10^3$ using repeated multiplication.

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Kits, cargo fractions, and space calculations use factors, fractions, and exponents correctly. Goes beyond the minimum with a creative, realistic choice.	Kits, cargo fractions, and space calculations use factors, fractions, and exponents correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Manifest includes kit contents, cargo fractions, calculations, and a mission fact sheet. Polished, organized, and visually engaging.	Manifest includes kit contents, cargo fractions, calculations, and a mission fact sheet. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.